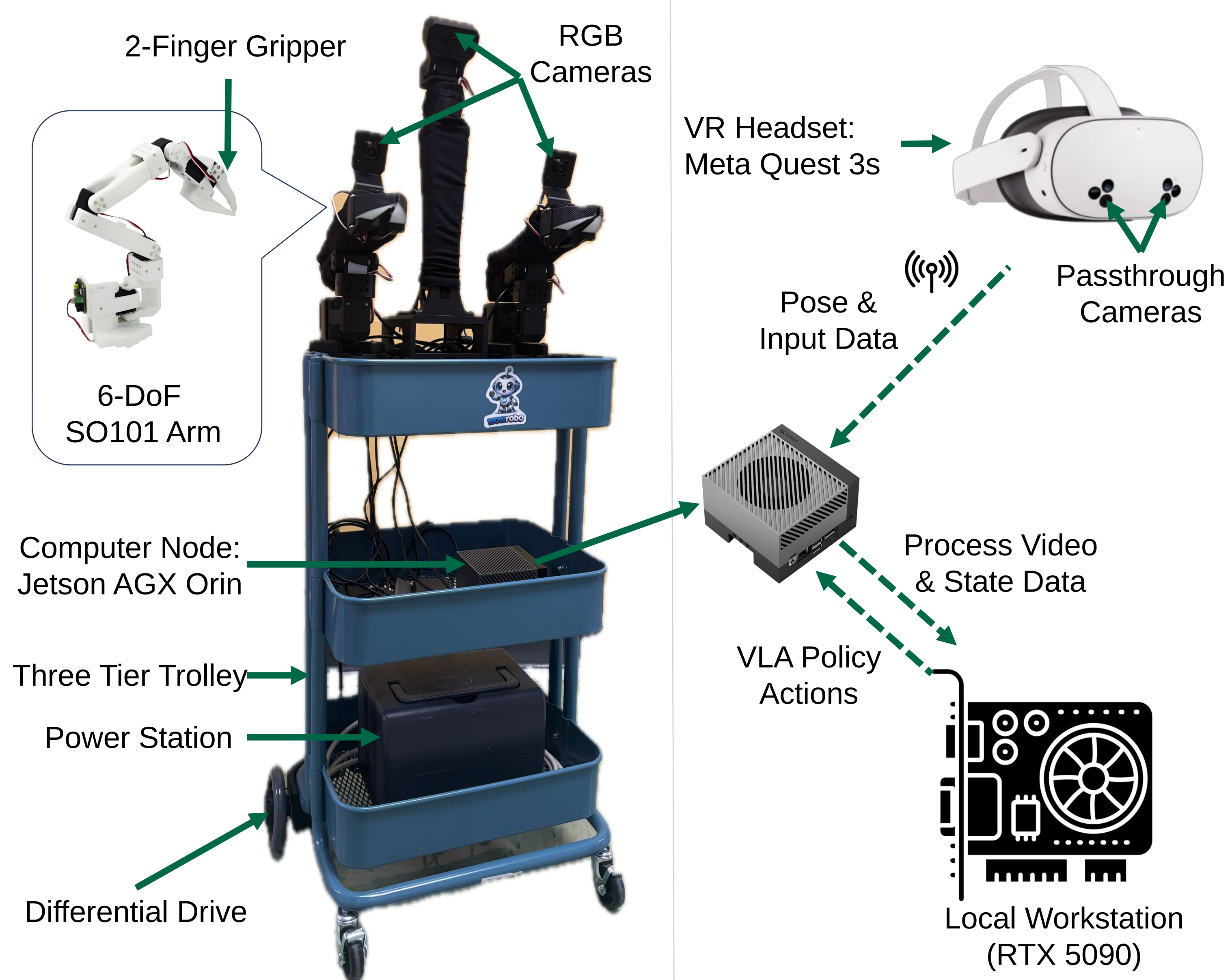


Introduction

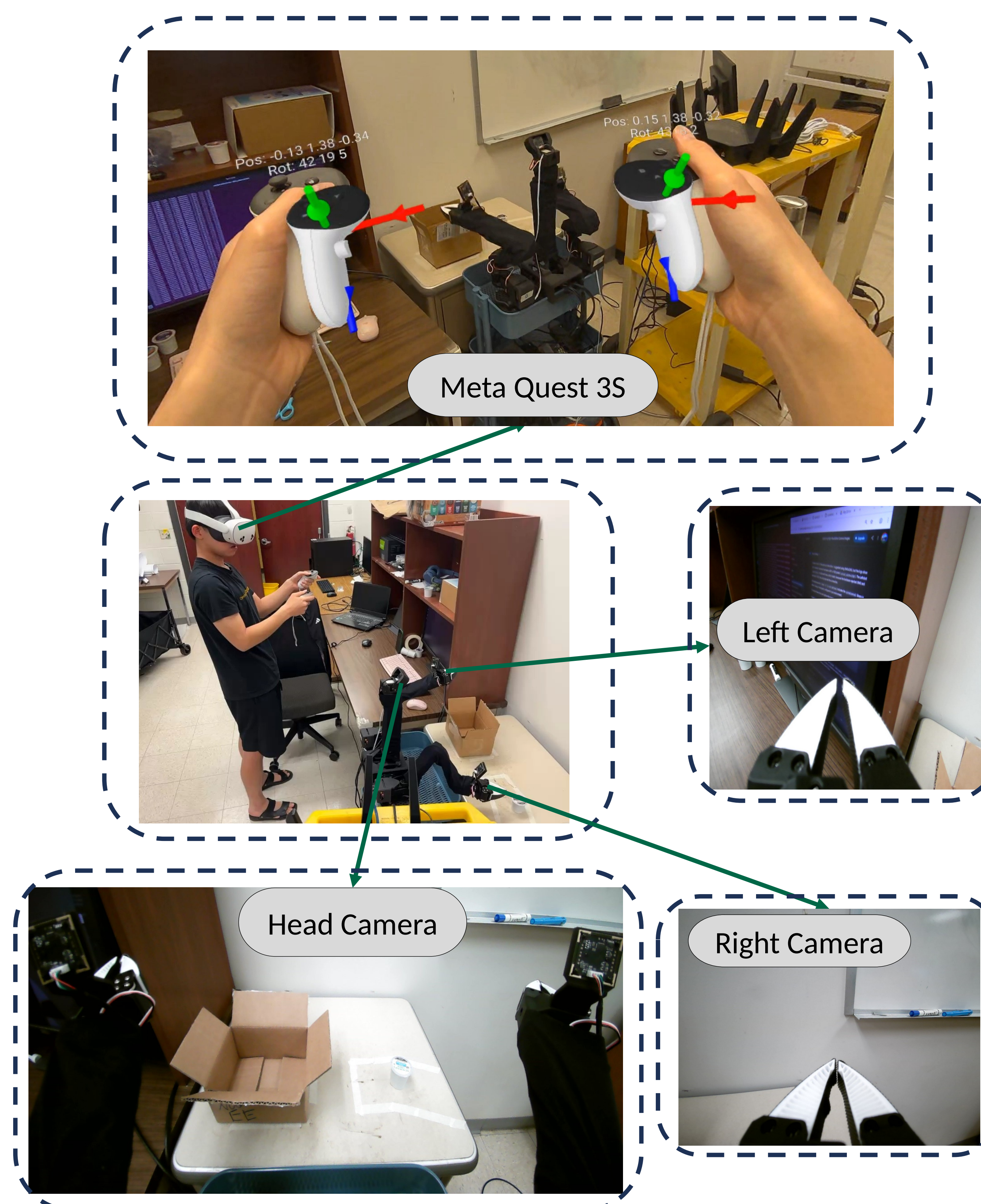
Although Vision Language Action models are advancing rapidly, high hardware costs restrict data collection, and these models often fail in unfamiliar environments without extensive retraining. To overcome these barriers, we propose a synergetic robotic system. First, we utilize the XLeRobot [1], an affordable open-source mobile platform with distributed computing. Next, we establish policy baselines using Meta Quest AR teleoperation and the Action Chunking with Transformers algorithm. Finally, we introduce a Robotic Retrieval Augmented Generation framework. By leveraging a hierarchical vector database, this system uses past experiences to prompt the model and enables immediate adaptation to novel tasks.

Hardware

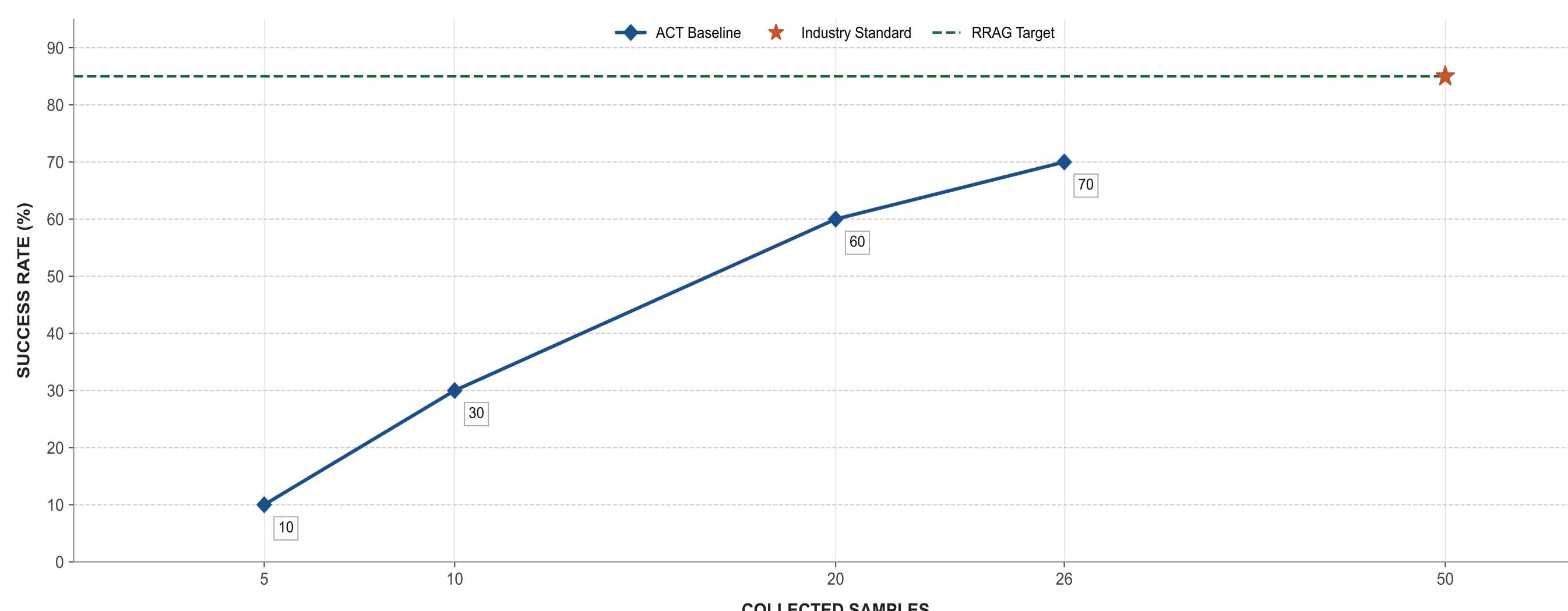
ONBOARD HARDWARE COMPUTE & CONTROL



Data Collection

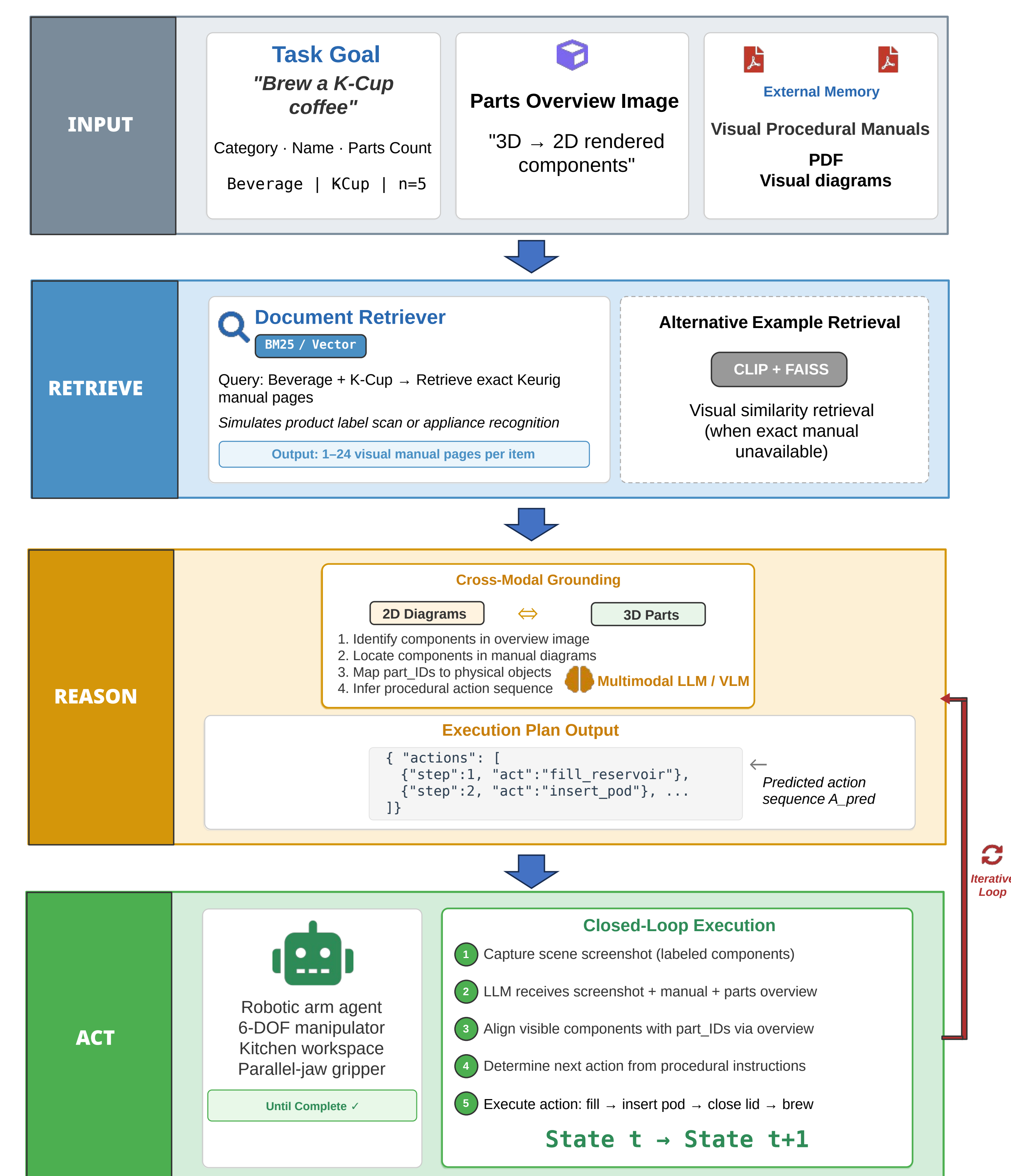


Data Collection



We collected 26 AR-headset demonstrations of picking up a K-cup from a table and placing it into a designated box. We trained an Action Chunking Transformer (ACT) with varying sample sizes. The figure shows the inference success rate after 10,000 steps across these settings.

Methodology



Future work

We will implement the proposed Robotic Retrieval Augmented Generation framework on the XLeRobot platform to measure generalization across novel tasks. Evaluation is based on metrics like success rates, inference latency, and recovery using unfamiliar objects and compositional sequences.

References

[1] Gaotian Wang, Zhuoyi Lu, Huang Yiyang, and Liu Yihao. Xlerobot: Apractical low-cost household dual-arm mobile robot design for general manipulation. <https://github.com/Vector-Wangel/XLeRobot>, 2025.